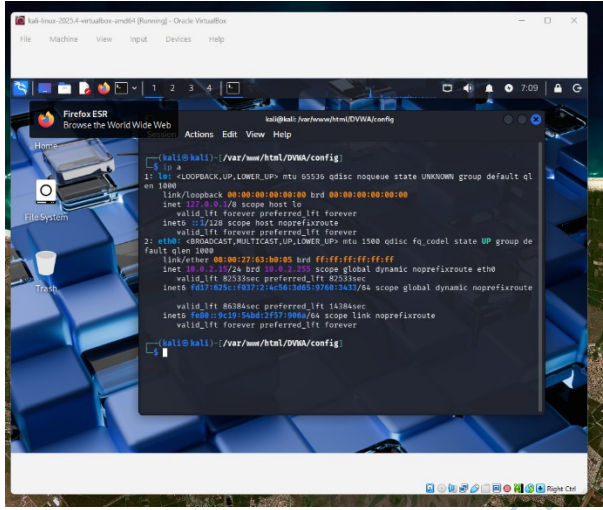


Practical 02

Important: This practical must be performed only in a controlled lab or virtual environment. Network scanning should be done only on authorized systems. Do not scan public or unknown networks.

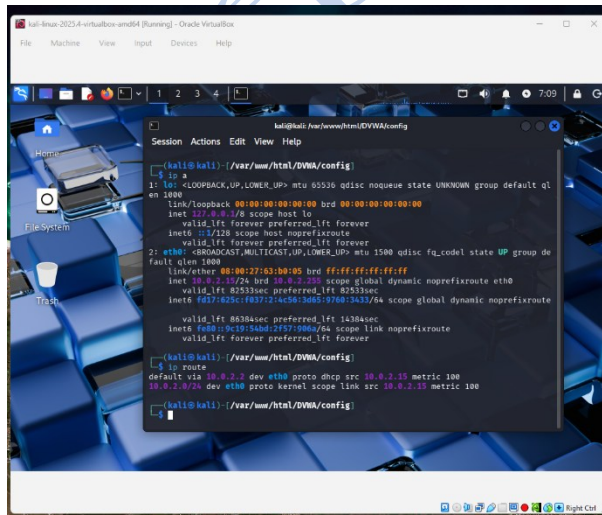
1. Open Terminal. Run the following command
 - a. `ip a` : Used to display IP address and network interface details.



Expected: IP address (e.g., 10.0.2.15) should be displayed.

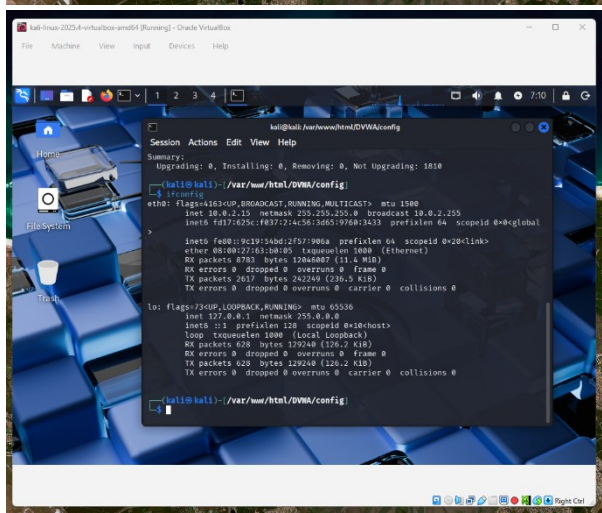
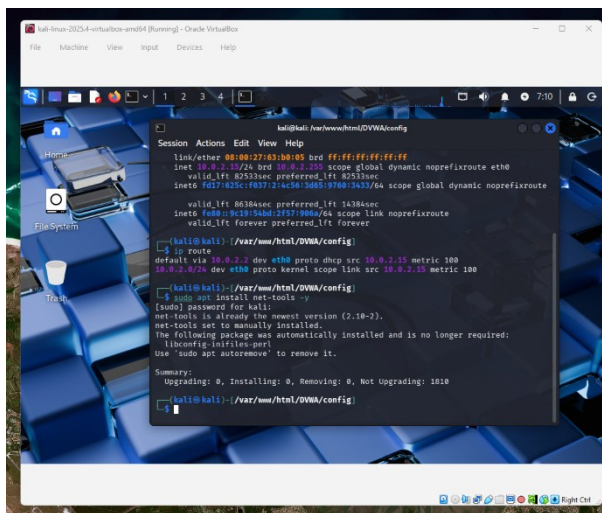
2. Check Default Gateway.
 - a. Run the following command:

`ip route`



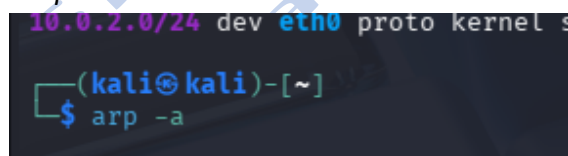
3. Running ifconfig: Run the following commands:
 - a. `sudo apt install net-tools -y`
 - b. `ifconfig`

Note: ifconfig is a legacy command used to view network configuration. It is not installed by default in modern systems.



4. View ARP Table

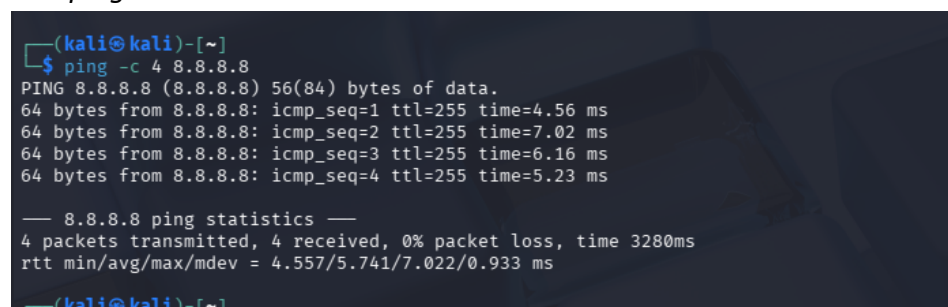
- Run `arp -a`



Note: If no entries are shown, first run the ping command and then execute `arp -a` again.

5. Test Connectivity.

- Run `ping -c 4 8.8.8.8`

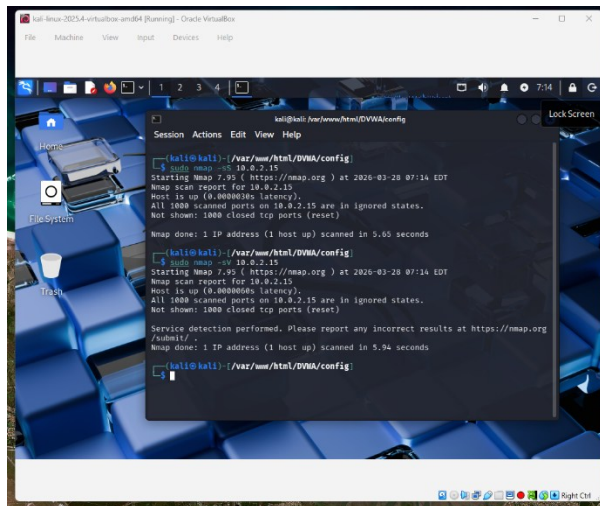


Expected: Replies should be received with 0% packet loss.

6. Scan Open Ports of Target System. Run the following commands:

- `sudo nmap -sS <target-ip>`
- `sudo nmap -sV <target-ip>`

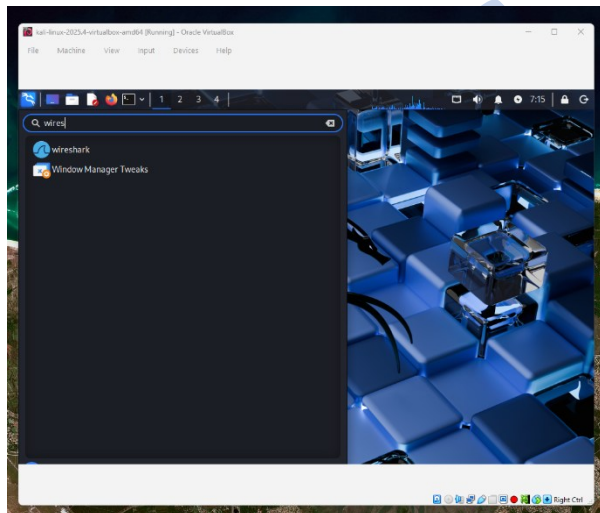
Replace <target-ip> with the IP address obtained using ip a.
Performs a TCP SYN scan to identify open ports.



Expected: List of open ports and services will be displayed.

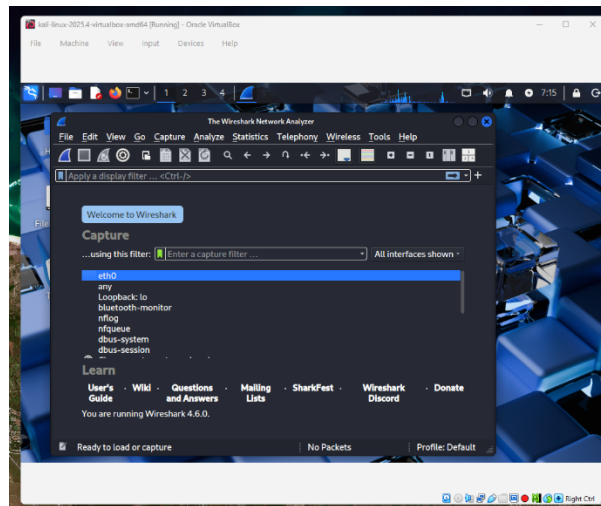
7. Packet Analysis

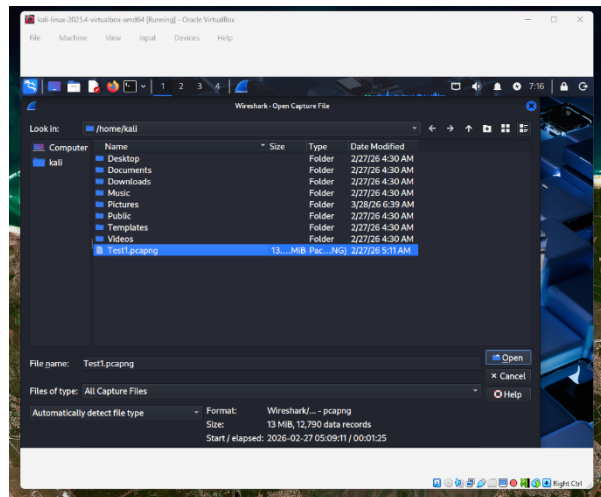
a. Open Wireshark



b. Open pcap File

- Click File → Open
- Select downloaded .pcap file



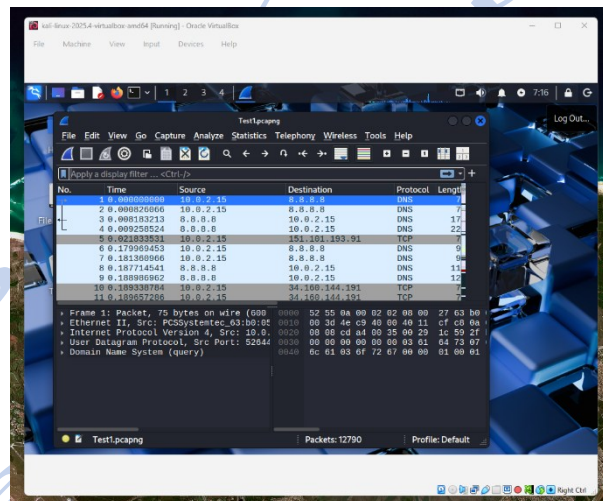


Note: Use a sample .pcap file downloaded from a trusted source

c. Analyze Traffic

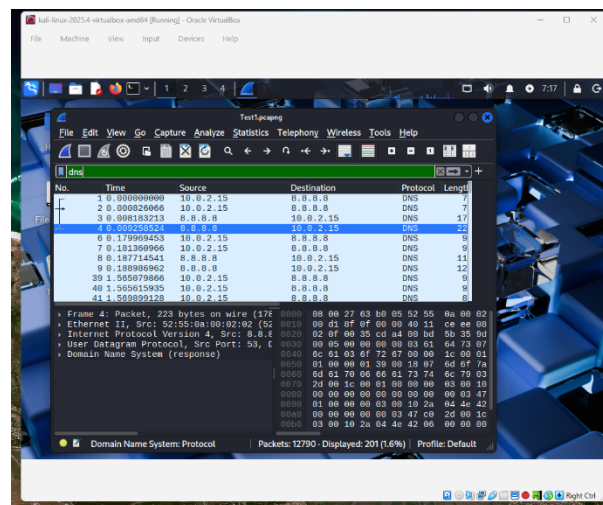
i. Look for:

1. TCP handshake (SYN, SYN-ACK, ACK)
2. DNS requests
3. HTTP packets



d. Use filter examples:

- i. tcp
- ii. dns
- iii. http



Expected: Network packets should be visible in the interface.